ONE DIRECTION AT A TIME

When A is diagonalizable, $\mathbf{x}' = A\mathbf{x}$ decouples into independent equations $y' = \lambda y$ with solutions $c e^{\lambda t}$. Each eigenvalue decides the behavior along its own eigenvector.

ALONG EIGENVECTOR $\mathbf{v}$

$$c e^{\lambda t} \mathbf{v}$$

split by **the eigenvalue λ**

↑

CASE 1

$\lambda > 0$

Exponential growth

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CASE 2

$\lambda < 0$

Decay

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CASE 3

$\lambda = 0$

Constant component

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CASE 4

λ complex

Oscillation

sines and cosines modulated by exponential envelopes; see **complex eigenvalues**

TAKEAWAY

The constants $c_1, \ldots, c_n$ come from the initial condition: write $\mathbf{x}(0)$ as a combination of the eigenvectors and read off the coefficients.